

LLMPirate Assignment Report

Course: LLM4ChipDesign

Instructors: Prof. Ramesh Karri

1. Overview

This report presents the implementation and evaluation of the LLMPirate workflow for rewriting hardware IP using Large Language Models. The objective of this assignment is to analyze whether LLM-based transformations can evade text similarity detection (SIM) while preserving the functional correctness of the circuit.

The study evaluates multiple rewriting strategies applied to the ISCAS-85 C432 benchmark (c432-CS320 variant). Each strategy is assessed based on:

- **Text Similarity (SIM)** — ability to evade detection
- **Functional Equivalence** — verified using Yosys formal checking

The assignment defines success as achieving **SIM < 0.3** while maintaining identical circuit behavior.

2. LLMPirate Workflow

The LLMPirate pipeline explores different strategies for rewriting a locked hardware design:

Input Circuit:

Gate-level Verilog netlist (topModule.v)

Processing Strategies:

- Direct Verilog rewriting using LLM
- Boolean equation rewriting → Verilog
- Divide-and-conquer (canonical mappings)
- Divide-and-conquer with LLM feedback
- Reference baseline (cached mappings)

Validation Steps:

- **Text Similarity (SIM):** Measures syntactic difference from the original
- **Formal Equivalence (Yosys):** Uses miter + SAT solving
 - Equivalent → PASS
 - Counterexample → FAIL

- Incomplete → UNKNOWN

This pipeline evaluates both **stealth (SIM)** and **correctness (logic equivalence)**.

3. Experimental Setup

- **Benchmark:** ISCAS-85 C432 (c432-CS320)
- **Circuit Type:** Combinational (top module, 212 gates)
- **Target Logic:** NOR-only implementation
- **LLM Used:** Gemini 2.5 Pro (via NYU gateway)
- **Equivalence Tool:** Yosys
- **Similarity Metric:** diffliB-based SIM (macOS fallback; rank-order remains meaningful, while Yosys is the authority for correctness)
- **Threshold:** SIM < 0.3

4. Results by Task

4.1 Task 1 — Direct Verilog Rewriting

Method:

Single LLM prompt to generate a full NOR-only Verilog module.

Observations:

- Output often incomplete (missing endmodule)
- Required multiple continuation attempts
- Parsing issues due to markdown/code formatting

Results:

- SIM $\approx$ 0.22 (passes threshold)
- Yosys: UNKNOWN (incomplete circuit)

Conclusion:

Low similarity was achieved, but functional correctness could not be verified.

4.2 Task 2 — Boolean Rewriting

Method:

Convert circuit → Boolean equations → LLM rewrite → reconstruct Verilog

Observations:

- Output was structurally complete

- Algebraic inconsistencies introduced errors

Results:

- SIM ≈ 0.13 (passes threshold)
- Yosys: NOT EQUIVALENT

Conclusion:

Produces valid syntax but incorrect logic behavior.

4.3 Task 3 — Divide-and-Conquer (Canonical NOR)

Method:

Replace each gate using deterministic NOR templates (no LLM guessing)

Observations:

- Fully systematic transformation
- No ambiguity in mappings

Results:

- SIM ≈ 0.015 (very low)
- Yosys: EQUIVALENT

Conclusion:

The only method achieving both stealth and correctness.

4.4 Task 4 — Divide-and-Conquer with Feedback

Method:

LLM generates gate mappings with iterative retry on format errors

Observations:

- Format improved after retries
- Semantic errors remained (invalid NOR structures)
- Some gates unmapped $\rightarrow$ fallback to original

Results:

- SIM ≈ 0.28 (passes threshold)
- Yosys: NOT EQUIVALENT

Conclusion:

Feedback improves formatting but does not guarantee correctness.

4.5 Task 5 — Reference Baseline**Method:**

Use cached mappings labeled *GPT3dot5* in the research pickle (not necessarily a fresh GPT-3.5 API call)

Observations:

- Incomplete coverage of gate types
- Partial rewriting leads to inconsistencies

Results:

- SIM ≈ 0.14 (passes threshold)
- Yosys: NOT EQUIVALENT

Conclusion:

Cached strategies are not robust across all designs.

5. Summary Table

Strategy	SIM	Yosys Result	Meets Criteria
Task 1 — Direct	~0.22	Unknown	No
Task 2 — Boolean	~0.13	Not Equivalent	No
Task 3 — Canonical D&C	~0.01 5	Equivalent	Yes

Task 4 — D&C + Feedback	~0.28	Not Equivalent	No
Task 5 — Reference	~0.14	Not Equivalent	No

6. Key Insights

- **SIM alone is misleading:**
Multiple approaches achieved low SIM but failed functional correctness.
- **Structured transformations outperform LLM generation:**
Deterministic mappings (Task 3) avoid hallucination errors.
- **LLM limitations:**
 - Struggles with large structured outputs
 - Produces syntactically valid but logically incorrect designs
 - Feedback loops fix format, not semantics
- **Verification is critical:**
Formal equivalence checking (Yosys) is essential to validate results.

7. Conclusion

The LLMPirate assignment demonstrates that evading similarity detection is easy, but preserving correctness is difficult with LLM-based approaches.

Among all strategies, only the divide-and-conquer with canonical mappings successfully achieved both:

- Low similarity ($SIM < 0.3$)
- Functional equivalence

This highlights that LLMs alone are insufficient for reliable hardware rewriting without structured constraints and formal verification.

8. Reproducibility and Integrity

- Experiments executed using the NYU-hosted LLM gateway
- macOS environment used diffliB SIM fallback
- All failed, incomplete, or unknown equivalence results are reported explicitly